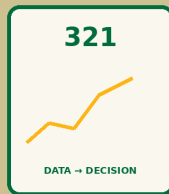
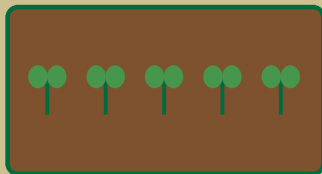


Formulate a Research Question



FORMULATE A RESEARCH QUESTION

TEST • MEASURE • ANALYZE • DECIDE

QUESTION → TEST → MEASURE → ANALYZE → MANAGE

PRACTICAL INVESTIGATION

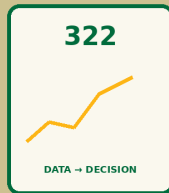
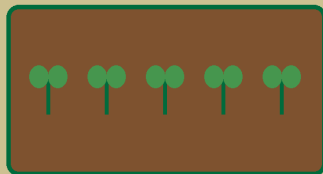
Carry out the investigation systematically, record the evidence, interpret the result, and state a defensible conclusion or management implication.

INTERPRET & DECIDE

What does the evidence support? State one limitation and the next measurement, test, or management action you would take.

DEFINE → MEASURE → RECORD → INTERPRET → DECIDE

Write a Hypothesis



WRITE A HYPOTHESIS

TEST • MEASURE • ANALYZE • DECIDE

QUESTION → TEST → MEASURE → ANALYZE → MANAGE

PRACTICAL INVESTIGATION

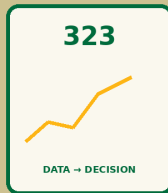
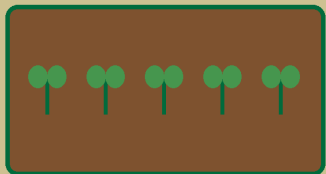
Carry out the investigation systematically, record the evidence, interpret the result, and state a defensible conclusion or management implication.

INTERPRET & DECIDE

What does the evidence support? State one limitation and the next measurement, test, or management action you would take.

DEFINE → MEASURE → RECORD → INTERPRET → DECIDE

Define Variables



DEFINE VARIABLES

TEST • MEASURE • ANALYZE • DECIDE

QUESTION → TEST → MEASURE → ANALYZE → MANAGE

PRACTICAL INVESTIGATION

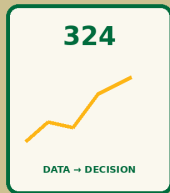
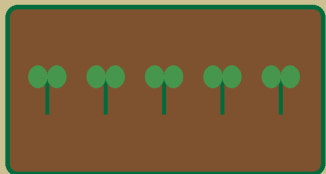
State the concept precisely, identify it in the garden system, and explain its relevance to the investigation or management decision.

INTERPRET & DECIDE

What does the evidence support? State one limitation and the next measurement, test, or management action you would take.

DEFINE → MEASURE → RECORD → INTERPRET → DECIDE

Choose a Control



CHOOSE A CONTROL

TEST • MEASURE • ANALYZE • DECIDE

QUESTION → TEST → MEASURE → ANALYZE → MANAGE

PRACTICAL INVESTIGATION

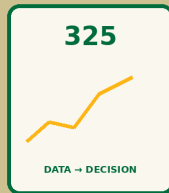
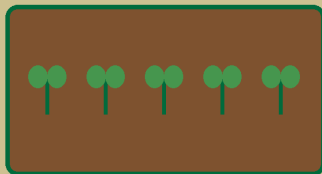
Carry out the investigation systematically, record the evidence, interpret the result, and state a defensible conclusion or management implication.

INTERPRET & DECIDE

What does the evidence support? State one limitation and the next measurement, test, or management action you would take.

DEFINE → MEASURE → RECORD → INTERPRET → DECIDE

Plan Replication



PLAN REPLICATION

TEST • MEASURE • ANALYZE • DECIDE

QUESTION → TEST → MEASURE → ANALYZE → MANAGE

PRACTICAL INVESTIGATION

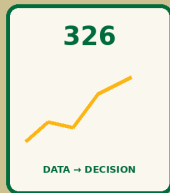
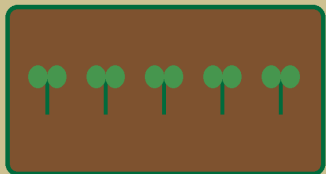
Define the objective and constraints, then prepare a practical layout, sequence, protocol, or rule that can be tested with garden evidence.

INTERPRET & DECIDE

What does the evidence support? State one limitation and the next measurement, test, or management action you would take.

DEFINE → MEASURE → RECORD → INTERPRET → DECIDE

Randomize Treatments



RANDOMIZE TREATMENTS

TEST • MEASURE • ANALYZE • DECIDE

QUESTION → TEST → MEASURE → ANALYZE → MANAGE

PRACTICAL INVESTIGATION

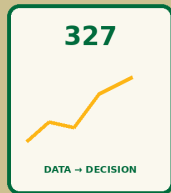
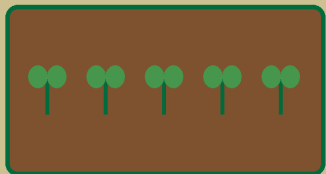
Carry out the investigation systematically, record the evidence, interpret the result, and state a defensible conclusion or management implication.

INTERPRET & DECIDE

What does the evidence support? State one limitation and the next measurement, test, or management action you would take.

DEFINE → MEASURE → RECORD → INTERPRET → DECIDE

Develop a Measurement Protocol



DEVELOP A MEASUREMENT PROTOCOL

TEST • MEASURE • ANALYZE • DECIDE

QUESTION → TEST → MEASURE → ANALYZE → MANAGE

PA-327 • Develop a Measurement Protocol

PRACTICAL INVESTIGATION

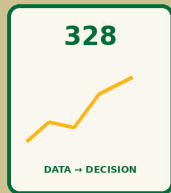
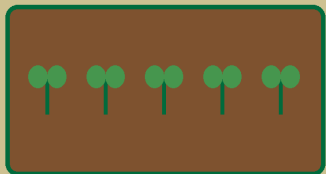
Select the appropriate instrument or method, follow a consistent protocol, record units, and repeat measurements where appropriate.

INTERPRET & DECIDE

What does the evidence support? State one limitation and the next measurement, test, or management action you would take.

DEFINE → MEASURE → RECORD → INTERPRET → DECIDE

Calculate Mean and Range



CALCULATE MEAN AND RANGE

TEST • MEASURE • ANALYZE • DECIDE

QUESTION → TEST → MEASURE → ANALYZE → MANAGE

PA-328 • Calculate Mean and Range

PRACTICAL INVESTIGATION

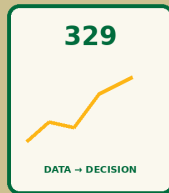
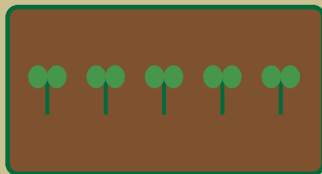
Use the recorded measurements and the appropriate equation. Show substitutions, units, result, and a brief interpretation.

INTERPRET & DECIDE

What does the evidence support? State one limitation and the next measurement, test, or management action you would take.

DEFINE → MEASURE → RECORD → INTERPRET → DECIDE

Graph Treatment Results



GRAPH TREATMENT RESULTS

TEST • MEASURE • ANALYZE • DECIDE

QUESTION → TEST → MEASURE → ANALYZE → MANAGE

PRACTICAL INVESTIGATION

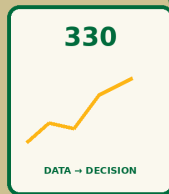
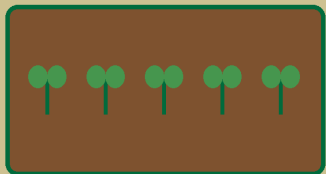
Carry out the investigation systematically, record the evidence, interpret the result, and state a defensible conclusion or management implication.

INTERPRET & DECIDE

What does the evidence support? State one limitation and the next measurement, test, or management action you would take.

DEFINE → MEASURE → RECORD → INTERPRET → DECIDE

Write an Experimental Conclusion



WRITE AN EXPERIMENTAL CONCLUSION

TEST • MEASURE • ANALYZE • DECIDE

QUESTION → TEST → MEASURE → ANALYZE → MANAGE

PRACTICAL INVESTIGATION

Carry out the investigation systematically, record the evidence, interpret the result, and state a defensible conclusion or management implication.

INTERPRET & DECIDE

What does the evidence support? State one limitation and the next measurement, test, or management action you would take.

DEFINE → MEASURE → RECORD → INTERPRET → DECIDE